

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<reasoning process here >

We are given :

- A 2 gram sample of rock yields 0.969 grams of Zn₂Fe(CN)₆.
- We are to find the percentage of Zn in the original rock sample.
- The reactions are :
 1. $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$
 2. $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$

So , each mole of Zn₂Fe(CN)₆ contains 2 moles of Zn .

We are given :

- Molecular weight of Zn₂Fe(CN)₆ = 342.6 g/mol
- Atomic weight of Zn = 65.4 g/mol so 2 Zn atoms = 2 × 65.4 = 130.8 g

Step 1 : Find moles of Zn₂Fe(CN)₆ produced :

moles = mass / molar mass = 0.969 g / 342.6 g/mol ≈ 0.002829 mol

Step 2 : Each mole of Zn₂Fe(CN)₆ comes from 2 moles of Zn .

So , moles of Zn = 2 × 0.002829 = 0.005658 mol

Step 3 : Mass of Zn = moles × atomic weight = 0.005658 × 65.4 ≈ ?

Compute :
0.005658 × 65.4 =
First: 0.005 × 65.4 = 0.327
0.000658 × 65.4 ≈ 0.0429
Total ≈ 0.327 + 0.0429 = 0.3699 g

So , mass of Zn = 0.3699 g

Step 4 : Percentage of Zn in the 2 g sample :

(0.3699 / 2) × 100 = 18.495% ≈ 18.5%

Close st option is 18%

Answer is F . 18% Zn

</reasoning process>
<answer>F</answer>